



ON-SITE360
construction software solutions

PROJECT PROPOSAL



GROUP 13

SCS 3214 / IS 3113: Group Project II - 2025

Project Proposal

Proposed Project Title: OnSite360

Project Group Details

Group number: 13

Group members:

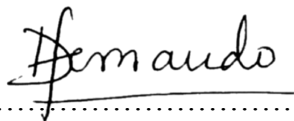
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Details of Project Supervisor, Co-supervisor, Advisors and Clients

Proposed Project Supervisor (Academic Staff of UCSC):

Name of the supervisor: Dr. Dinuni K Fernando

Signature of the supervisor:



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Date: 5/30/2025

Proposed Project Co-Supervisor (Assigned by Course Coordinator):

Name of the co-supervisor: Ms. T.U.S Wijewardhana

Signature of the co-supervisor:



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Date: 5/30/2025

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1. Project Title

OnSite360 - Construction Project Management System

2. The Goal and Objectives

2.1 Project Goal

OnSite360 is a mobile-first construction project management platform designed to unify communication, documentation, and workflows across all levels of a construction organization, from field workers to executives. Most construction projects suffer from fragmented communication, delayed decision-making, and inefficient data flow, resulting in cost overruns and missed deadlines. OnSite360 bridges these gaps by:

- Connecting the construction site with the office in real time, eliminating delays caused by emails, calls, and manual reporting.
- Digitizing and automating workflows, reducing paperwork and human errors.
- Providing data-driven insights, helping managers and executives make faster, better-informed decisions.

The ultimate goal of OnSite360 is to reduce project delays, lower administrative costs, and enhance overall construction efficiency.

2.2 Our Objectives

1. Real-Time Communication

Construction teams waste days waiting for clarifications (e.g., design changes, material approvals), resulting in idle labor and schedule delays. As a solution, OnSite360 provides an integrated chat system that connects site supervisors, engineers, and managers instantly, with:

- Threaded discussions are tied to specific tasks, drawings, or Requests for Information (RFIs).
- File attachments (photos, markups, documents) for faster decision-making.

So RFI resolution time drops from days to hours, keeping projects on track.

2. Centralized Data Access

Many critical project data (daily logs, progress reports, inspection records) are scattered across emails, spreadsheets, and paper logs, making it hard to track. We provide a cloud-based platform where:

- Site supervisors log progress, issues, and photos via mobile (even offline).
- Engineers & managers access real-time, structured data for approvals and adjustments.

This eliminates data silos, ensuring all stakeholders work from a single source of truth.

3. Project Transparency

Custom executive dashboards showing:

- Progress vs. timeline (Gantt comparisons).
- Labor productivity & cost trends.
- Safety/compliance alerts.

This improves proactive decision-making to prevent budget overruns and delays.

4. Process Automation

Manual processes (e.g., task assignments, document approvals, and daily reporting) slow down operations. Our system automates features like:

- Automated task routing (e.g., auto-assigning RFIs to the right engineer).
- Automated reminders for overdue inspections or approvals.
- Report generation (e.g., daily logs → executive summaries).

This reduces administrative workload by 30-50%, freeing teams for high-value work.

5. Scalability

Many construction tech tools are too complex or expensive for mid-sized firms. But our system is a lean, intuitive platform with:

- Role-based interfaces (simplified for field crews, detailed for managers).
- Minimal training (adoption within days, not weeks).

This enables small to mid-sized firms to compete with enterprise-grade tools.

3 . Tentative Problem Definition

Construction projects frequently face delays and cost overruns due to:

- **Communication Gaps**

In traditional construction environments, communication between on-site teams and off-site decision-makers is often slow, fragmented, and reliant on outdated channels such as email, phone calls, or printed documents. For example, if a site supervisor

encounters a design ambiguity or clash during execution, they must typically raise a Request for Information (RFI) via email or form, wait for it to be seen by a project manager or architect, and then wait again for a formal response. These delays, often spanning **several days**, result in **idle labor**, **missed productivity targets**, and **disruptions to project timelines**. The ripple effect of one unresolved clarification can impact scheduling, crew assignments, material usage, and even inspections. OnSite360 addresses this problem by providing a **real-time chat system linked to tasks and documents**, ensuring that field teams can instantly raise issues and decision-makers can respond from anywhere, reducing downtime and improving accountability.

- **Fragmented Data**

In many existing systems, **site logs**, **material delivery records**, **labor attendance**, **progress photos**, **safety inspections**, and **official project documents** are managed using separate tools or even paper-based methods. These records are often siloed across spreadsheets, emails, PDF folders, and printed forms, making it extremely difficult for decision-makers to obtain a consolidated view of what is happening on the ground. This fragmentation hinders real-time oversight, slows **progress tracking**, **report generation**, **issue detection**, and **compliance auditing**. It also increases the risk of miscommunication, data loss, and duplicated effort. OnSite360 addresses this by centralizing all critical construction data, progress updates, photos, crew logs, safety reports, and documents into a **single, unified platform** accessible via mobile or web, with offline sync support. This ensures better visibility, traceability, and faster decision-making.

- **Inefficient Workflows**

Despite the digital age, many construction sites still rely heavily on **manual, paper-based workflows**. Daily logs are handwritten, checklists are filled out on printed forms, and critical documents (like inspection approvals or RFIs) are emailed back and forth between multiple stakeholders. This results in **delays**, **missing data**, **inconsistent formats**, and **difficulty in tracking historical actions**. Paper checklists are easy to lose, email chains get buried, and approvals take too long to circulate, creating bottlenecks that cost both time and money. OnSite360 replaces these inefficient workflows with **digital checklists**, **mobile-first logs**, **auto-synced updates**, and **smart task automation**. The system not only streamlines execution but also enables proactive alerts, task assignments, and report generation, eliminating redundancy and speeding up operations.

OnSite360 addresses these by syncing field teams[1] with office stakeholders through a unified platform, ensuring faster decisions and streamlined operations.

4. A Brief Introduction To The Project:

Our project aims to address key challenges faced in the construction industry by developing a smart decision-support system tailored for engineers, project managers, executives, and on-site workers. The idea originated after a discussion with a civil engineer who actively works on construction projects. He highlighted several common issues encountered during

project execution, including communication gaps, inefficient data handling, delays, and a lack of real-time decision-making support.

Motivated by these insights, we set out to create a solution that simplifies and accelerates the decision-making process on construction sites. The proposed system processes and presents relevant data in a clear and accessible format, enabling engineers to make faster and more informed decisions. Project managers can maintain workflow efficiency and minimize delays, while executives benefit from cost-effective project execution. Furthermore, workers and supervisors are supported in maintaining consistent progress without unnecessary interruptions.

5. The Scope Of The Project:

The aim of OnSite360 is to deliver a modern, mobile-first construction management platform that streamlines Pre-Construction document control, Workforce and On-Site Productivity, and Project Execution Management, with a leaner and more intuitive user experience.

OnSite360 will reduce administrative overhead, enhance field efficiency, and offer intelligent automation that mid-sized firms can afford, adopt, and scale with minimal training.

5.1 Users Of The System

- 1. System Admin**

User management, security, integrations, and technical configurations.

- 2. Executive Admins**

Cross-project analytics, policy compliance, and strategic oversight.

- 3. Project Directors**

Multi-project governance, milestone tracking, and document approval.

- 4. Project Manager**

Task scheduling, workforce allocation, RFI/document coordination.

- 5. Engineers/Architects**

Drawing reviews, RFI responses, and technical compliance sign-offs.

- 6. Site Supervisors**

Daily logs, crew assignments, safety/QA checklists, and real-time issue reporting.

- 7. Sub Contractors**

Progress updates, issue logging, and limited document access.

- 8. Clients**

Optional access to view project dashboards and documents, including milestone tracking and report exports.

5.2 Main Functionalities Of The System

5.2.1. Executive Admin

Provides **business oversight** across the organization and all projects. Focuses on **project performance**, **organizational analytics**, and **strategic decisions**.

5.2.1.1. Cross-Project Dashboards & Organizational Analytics

- View real-time performance across all projects:
 - Progress % by project
 - Labor productivity and efficiency trends
 - Safety incidents and compliance stats
- Drill-down dashboards:
 - Region → Project → Team → Individual
- Export high-level reports for board meetings or investors

5.2.1.2. Company Policies (Business-Level)

- Define and review:
 - Document standards
 - Field log protocols
 - Safety checklist frequency
- Approve project-level customization of business policies.
- Audit adherence to company-wide standards (non-technical)

5.2.1.3. Project Oversight

- Assign or reassign Project Directors and Managers
- View project portfolios and phase timelines
- Review milestone statuses, delays, and high-risk alerts
- Override project settings in business emergencies

5.2.1.4. Read-Only Audit Oversight

- View summaries of:
 - System usage trends
 - Compliance reports

5.2.2. System Administrator / IT Admin

Handles **technical configuration**, **system integrity**, **security**, and **infrastructure-level control**. Supports the platform from a technical and operational perspective, distinct from project or business decision-making.

5.2.2.1. User Role & Permission Management

- Create, edit, and delete **custom user roles**.
- Define **granular module-level permissions** (e.g., who can view RFIs, edit calendars, manage documents)
- Manage **bulk role assignments** and department-based permissions.

5.2.2.2. User Management

- Add, deactivate, or transfer users across projects
- Set user-level limits (e.g., file sizes, access timeframes)
- Monitor user login activity and anomalies

5.2.2.3. System Logs & Audit Trail

- View and export:
 - Authentication logs
 - API usage and errors
 - File access and permission changes
- Monitor system health, uptime, and usage metrics to ensure optimal performance.

5.2.2.4. Integrations & APIs

- Manage third-party integrations
- Configure and rotate API keys

5.2.2.5. Infrastructure & Maintenance

- Coordinate system updates and version rollouts
- Oversee backup policies and restore points
- Ensure data security protocols (encryption, storage compliance)
- Handle tech support escalations and system-level tickets

5.2.3. Project Director

The Project Director is responsible for overseeing multiple construction projects under the organization. They serve as the operational bridge between Executive Admins and Project Managers, ensuring the proper setup, consistency, performance, and compliance across all assigned projects.

5.2.3.1. Project Creation and Initialization

- Create and initialize new projects from scratch or using **company-approved templates**.
- Define core project parameters, including type, location, size, start and end dates, and budget reference.

- Select applicable **compliance policies** and checklists set by the Executive Admin.
- Select default roles and permissions for each project type.

5.2.3.2. Project Team Structuring

- Assign **Project Managers, Engineers, Architects, Supervisors, and Subcontractors** to projects.
- Control or delegate user role invitations at the project level
- Modify project-specific roles (e.g., Quality Assurance Specialist, Subcontractor Project Manager)
- Define communication groups (e.g., Design Team, Field Ops, Safety Leads)

5.2.3.3. Multi-Project Oversight

- View cross-project dashboards:
 - Progress % completion
 - Delay heatmaps
 - Safety compliance metrics
 - Productivity vs estimates
- Compare project performance regionally or by type (e.g., Residential vs Commercial)

5.2.3.4. Documentation Oversight

- Approve final versions of project documents before execution (drawings, contracts)
- Ensure project documents are correctly tagged and versioned
- Oversee submittals, RFIs, and spec-related communications
- Assign responsibilities for document review cycles

5.2.3.5. Project Alerts & Issue Escalations

- Set thresholds for delays, cost overruns, or log inconsistencies
- Receive alerts for non-compliance or overdue milestones
- Intervene directly or escalate issues to the Executive Admin

5.2.3.6. Milestone & Timeline Governance

- Define master milestones for each project
- Approve changes to baseline timelines
- Compare Gantt schedules vs actual progress across multiple projects

5.2.3.7. Audit & Performance Reviews

- View user activity logs within assigned projects
- Run performance reviews of PMs and Site Supervisors
- Provide input on team reallocation based on performance

5.2.4. Project Manager (PM)

The Project Manager is responsible for **executing the project plan**, coordinating with all stakeholders, **managing day-to-day progress**, and ensuring the project stays on **schedule, within budget, and compliant** with quality and safety standards.

5.2.4.1. Task & Schedule Management

- Create, assign, and update **tasks and subtasks** with dependencies
- Adjust schedules in a **drag-and-drop Gantt chart**
- Set task priorities and milestone deadlines
- Monitor real-time progress updates from site supervisors
- Get alerts for overdue or off-schedule tasks

5.2.4.2. RFI, Submittal & Document Coordination

- Create and track **RFIs** and submittals with due dates
- Route documents to engineers or architects for review
- Maintain revision history and attach linked drawings/specs
- Comment and clarify design questions within the platform
- Approve or escalate unresolved RFIs

5.2.4.3. Workforce & Crew Oversight

- Assign crews or individual workers to tasks based on availability and skill tags.
- Monitor daily attendance, work hours, and productivity rates
- Reallocate resources across tasks based on progress or delays

5.2.4.4. Calendar & Daily Log Management

- Maintain a **centralized project calendar**
- Review and approve **daily logs** submitted by site supervisors:
 - Weather
 - Labor
 - Material usage
 - Site photos
- Attach notes or corrective actions to logs.

5.2.4.5. Progress & Performance Tracking

- View dashboards comparing **planned vs actual** work
- Analyze **productivity KPIs**: labor hours, material usage, delays
- Download reports or auto-share weekly summaries with directors
- Flag and explain deviations from the baseline timeline

5.2.4.6. Checklist & Quality Control

- Assign Quality Assurance (QA) / Quality Control (QC) or safety checklists to site teams
- Review inspection results and corrective action logs
- Ensure compliance with project-level standards
- Validate readiness for the next construction phase

5.2.4.7. Communication & Collaboration

- Chat with team members via **project-specific threads**
- Attach documents, photos, and markups in chats
- Communicate with clients or directors through official updates
- Schedule internal or stakeholder review meetings

5.2.5. Site Supervisor

The Site Supervisor is responsible for **coordinating day-to-day fieldwork**, **supervising the team**, **enforcing safety protocols**, and **providing real-time reporting** from the job site. They are the key communicator between the site and the office (PMs, Engineers).

5.2.5.1. Daily Logs Management

- Submit detailed **daily site reports** including:
 - Weather conditions
 - Headcount (by crew/trade)
 - Work completed
 - Site conditions & delays
- Attach **photos, videos, and files** to logs
- Tag specific tasks or crews for reference

5.2.5.2. Task & Crew Assignment

- View and manage **assigned tasks from PM**
- Allocate tasks to **specific workers or subcontractor crews**
- Track **task status** (Not Started, In Progress, Completed)
- Adjust assignments based on crew availability or delays

5.2.5.3. Photo & Media Capture

- Upload **geotagged progress photos** and videos
- Categorize images by task, date, crew, or issue
- Annotate images if needed (e.g., highlight problem areas)

5.2.5.4. Schedule Tracking & Check-ins

- View the project calendar and current schedule from Gantt
- Mark task progress on the field in real-time
- Conduct **start-of-day briefings** and record attendance

5.2.5.5. Checklist Compliance

- Complete and submit:
 - **Safety checklists** [2] (PPE, equipment use, site hazards) [2]
 - **QA/QC checklists** (material quality, finish checks)
 - **Inspection readiness** reports
- Flag issues or submit **non-compliance reports**

5.2.5.6. Issue Reporting & Field Communication

- Report issues (equipment failure, material shortage, design mismatch)
- Tag relevant teams (e.g., PM, Architect) for escalation
- Use the **chat feature** to collaborate with the office
- Record **notes or quick field updates**

Material & Equipment Monitoring

- Log material deliveries and usage per task
- Monitor equipment usage, downtime, or shortages
- Submit **equipment inspection checklists**

5.2.6. Engineer / Architect

Ensure the technical accuracy, design integrity, and regulatory compliance of construction projects. Review RFIs, approve drawings, and provide timely responses to field and design queries.

5.2.6.1. RFI (Request for Information) Management

- Review and respond to RFIs submitted by PMs or Site Supervisors
- Add annotations, attach clarifications, and approve or escalate
- Track status (Open, In Review, Answered, Closed)
- View full audit trail of responses and document revisions

5.2.6.2. Drawing & Document Review

- Upload and manage project drawings and plans
- Use version control to track changes and updates
- Markup documents for revisions, comments, or approvals
- Approve drawings for construction or return with comments

5.2.6.3. Submittals & Specifications Oversight

- Review material submittals, shop drawings, and specs
- Approve or reject submittals with a reason
- Link submittals to relevant RFIs or drawings for traceability

5.2.6.4. Technical Compliance & Inspection Support

- Review QA/QC checklists completed by site teams
- Provide sign-off for completed stages [3] (e.g., structural, MEP)
- Schedule or request field inspections when needed
- Issue field directives (e.g., redesign details, alternate methods)

5.2.6.5. Knowledge Base & Standards Contribution

- Upload design guides, best practices, and reference materials
- Standardize document templates for drawings, RFIs, and submittals
- Maintain FAQs for typical design or construction questions

5.2.6.6. Collaboration & Coordination

- Communicate with PMs and supervisors about technical concerns
- Join scheduled reviews or coordination meetings via the platform
- Alert the team to drawing updates or spec changes

5.2.7. Subcontractor

External specialists or companies are hired to execute specific project scopes (e.g., electrical, plumbing). They are granted limited, task-specific access to update progress, report issues, interact with relevant documents, and communicate with the internal team, **without exposure to sensitive project or organizational controls**.

5.2.7.1. Progress Tracking

- Update assigned task status and report % completion
- Upload site progress photos with optional annotations
- Log hours worked per day, per task, or per crew

5.2.7.2. Issue Reporting

- Report on-site problems or incidents, attach photos, and auto-tag the location
- Mark hazards and provide comments for clarification
- Request field inspections or supervisor approvals as needed

5.2.7.3. Document Interaction

- View shared project documents: drawings, installation manuals, specifications.
- Fill and submit installation or inspection checklists
- Respond to RFIs, submittals, or field inquiries initiated by the project team

5.2.7.4. Daily Logs

- Fill in daily site activity: workforce count, equipment usage, and material delivered.
- Attach notes, delays, or on-site events (e.g., weather, safety alerts)

5.2.7.5. Communication

- Receive system or project-wide announcements from PMs and Supervisors
- Tag supervisors, PMs, or engineers in comments for collaborative issue resolution

5.2.7.6. Mobile-first Experience

- Use OnSite360 as a mobile Progressive Web App (PWA)
- Capture field data quickly using mobile camera, voice, or form inputs
- Offline-first functionality for data sync when the network is available

5.2.8. Client/ Stakeholder

Monitor progress, view deliverables, review reports, and stay informed about the project's health. Clients are **read-only observers** with access to key milestones and updates, without modifying the actual project data.

5.2.8.1. Project Dashboard Overview

- Access a **read-only executive dashboard** for real-time project status:
 - Overall % completion
 - Active vs. delayed tasks
 - Upcoming milestones
 - Budget vs. actual summaries (if permitted)
- Filter data by project, phase, or location

5.2.8.2. Milestone & Timeline Tracking

- View Gantt charts or milestone calendars
- See planned vs. actual timelines
- Get automatic notifications for:
 - Phase completions
 - Delay justifications
 - Key approvals

5.2.8.3. Visual Progress Monitoring

- Access site photos and videos tagged to tasks or dates
- Browse **before-and-after images**

5.2.8.4. Document & Drawing Access

- View finalized and approved:
 - Drawings
 - Specs
 - Reports
 - Safety documentation
- Download shared files

5.2.8.5. Report Review & Export

- Access pre-formatted:
 - Weekly/monthly progress reports
 - Cost summaries
 - Safety logs
- Export or share PDF reports directly with stakeholders

5.2.8.6. Stakeholder Communication

- Receive periodic updates or announcements from the PM/Director
- Comment or acknowledge reports and dashboards
- Subscribe to alerts for major events or delays

6. Tentative Technologies

The table below presents the technology stack planned for this project. These technologies were chosen based on their performance, scalability, and industry adoption. A more detailed justification for each can be found in [Section 11](#).

Category	Technology Stack
Environment & Deployment	Docker
Frontend Framework	Vite + React + PWA
Backend Framework	NestJS (Node.js)
Database	PostgreSQL
Version Control	Git & GitHub

Project Management	Jira
Diagramming	Draw.io
UI Design	Figma

7. Feasibility Study

7.1. Operational Feasibility

Our construction management platform, OnSite360, is designed to address the practical, field-level inefficiencies commonly faced by construction professionals across various roles, including engineers, project managers, site supervisors, and subcontractors.

In discussions with industry professionals, we identified several key operational challenges, including complex user interfaces, disconnected communication systems, limited access to field data, and a heavy reliance on external tools for collaboration, documentation, and reporting. Many existing systems, while feature-rich, are not aligned with day-to-day construction workflows and require significant training to operate effectively.

OnSite360 addresses these gaps through a simplified, mobile-first interface, real-time communication tools, custom role-based access control, and offline data collection support. Features such as smart task scheduling, field report automation, instant notifications, and permission-based document sharing provide a focused and intuitive experience for users at all levels.

By targeting core project management functions without the burden of unnecessary complexity, our system enables field teams, engineers, and administrative staff to operate more efficiently, thereby reducing downtime and improving decision-making speed. This approach makes OnSite360 highly operable and a practical fit for the construction sector.

7.2. Economic Feasibility

OnSite360 has been developed as a free and open-source solution, eliminating the significant licensing costs associated with commercial platforms like Procore, Buildertrend, and Autodesk BIM 360 [4]. These platforms often employ subscription-based pricing models that can be prohibitively expensive for small and medium-sized construction firms.

In contrast, our platform only requires standard deployment infrastructure, which can be hosted using cost-effective cloud solutions. Technologies like PostgreSQL, NestJS, Docker, and React are all open-source and maintained by large developer communities, removing the need for proprietary tools or paid software dependencies.

By keeping software, deployment, and hardware requirements minimal, OnSite360 ensures affordability for a wide range of users. This increases adoption potential and provides a higher return on investment for organizations that want to modernize without overspending.

7.3. Technical Feasibility

The technical foundation of OnSite360 has been carefully selected to ensure maintainability, modularity, and extensibility. The backend is built using NestJS, a Node.js framework that supports modular architecture and is well-suited for scalable enterprise-grade systems. The frontend is powered by React and Vite, allowing rapid interface development and performance optimization.

As a Progressive Web App (PWA), the platform supports offline usage, which is critical for field operations in low-connectivity areas. The use of PostgreSQL as the database enables handling both structured and semi-structured data, which is ideal for storing documents, logs, and various types of project metadata.

The system also leverages Docker for containerization, which simplifies deployment and environment management across local and cloud servers. All selected tools and libraries are actively supported by global developer communities, ensuring longevity and accessibility.

Overall, OnSite360 can be developed, maintained, and expanded by generalist full-stack developers, making the technical feasibility high even for smaller teams or academic groups.

7.4. Schedule Feasibility

The scope of OnSite360 has been structured into clear, manageable modules that align with common construction workflows. These modules include document management, task scheduling, chat and RFI systems, role management, inventory tracking, and reporting tools. Each module can be developed in parallel where needed, and their interdependence has been minimized to support waterfall development.

The total estimated development time is 15–16 weeks, which fits within the project duration. The workload distribution among team members makes the timeline achievable without overextension.

7.4.1. Human Resources Required:

- **Frontend Developer(s):** For building the user interface with React and Vite
- **Backend Developer(s):** For API development and database logic using NestJS and PostgreSQL
- **UI/UX Designer:** For designing user-friendly, responsive, and intuitive workflows
- **Project Analyst/Coordinator:** For planning, testing, and overseeing module integration
- **DevOps Engineer (basic level):** For Docker setup, testing environments, and deployment

7.4.2. Technical Resources Required:

- Development tools (VS Code, Git)
- Cloud hosting
- Docker containers for environment consistency
- PostgreSQL database server
- Browsers and mobile devices for PWA testing

7.4.3. Feasibility of Resource Acquisition:

All required tools and technologies are open-source and accessible. Human resource roles can be fulfilled by our group members. Hosting can be achieved using free-tier cloud services or minimal-cost student plans. No high-end hardware or specialized equipment is necessary.

Activity	Estimated Hours
Requirement Analysis & Planning	20
UI/UX Design	30
Frontend Development	120
Backend Development	120
API Integration & Role System	30
Testing & Debugging	40
Deployment & Optimization	20
Documentation & Report Writing	20
Contingency Buffer	30
Total Estimated Man-Hours	430 Hours

This estimate aligns with typical 3rd-year project credits and justifies team involvement across multiple roles over the semester-length timeline.

8. Main Deliverables Of The System

- Complete working software and source code.
- Complete Software Requirement Specification
- User manual
- Administrator's manual with deployment instructions

9. The Project Plan

Phase	Activity	May				June				July				August				September			
		W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
Problem Identification	Requirement Gathering																				
	Requirement Analysis																				
	Scope Identification																				
	Feasibility Study																				
System Analysis	System Architecture																				
	SRS																				
System Design	Database Design																				
	User Interface Design																				
System Development and Testing	Frontend Development																				
	Backend Development																				
	System Testing																				
Deployment	Deployment																				

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11. Technology Justification

11.1. Docker – *Environment Setup & Deployment*

Why Docker?

- Enables **consistent deployment environments** across dev, staging, and production.
- Simplifies onboarding for developers, with one command to set up the environment.
- Scales easily with **Docker Compose** or container orchestration tools like Kubernetes if needed.

Compared to alternatives (e.g., manual setup, VM-based environments):

- Faster setup, **less prone to "it works on my machine" issues**.
- **Lightweight** and supports rapid iteration.

Benefits:

- **Client:** Reduced deployment time and fewer environment-related issues.
- **Maintenance:** Easier to manage dependencies and isolate services for debugging or updates.

11.2. Vite + React + PWA – *Frontend Framework & Cross-Platform Access*

Why React + Vite?

- React is a **component-driven framework** that is perfect for systems like OnSite360, where multiple screens share common patterns.

- **Vite** is a next-generation frontend tool that offers:
 - Lightning-fast cold starts
 - Optimized hot module replacement
 - Better performance in dev and prod environments

Why PWA (Progressive Web App)?

- Enables the **web app to be installed like a native app** on iOS, Android, and desktop.
- Works **offline**, can send **push notifications**, and feels like a mobile app.
- **Avoids the cost of building and maintaining separate mobile apps** (iOS/Android).

Compared to alternatives:

- Compared to Angular or Vue, **React has a larger ecosystem and community**, making it easier to hire talent and extend its capabilities.
- Compared to React Native, the PWA approach **avoids duplicated codebases**.

Benefits:

- **Client:** Reduces development and long-term costs; provides a consistent user experience across devices.
- **Maintenance:** Reusable components, fast dev cycle, scalable codebase.

11.3. NestJS (Node.js Framework) – *Backend Framework*

Why NestJS?

- A **TypeScript-first**, structured framework for [Node.js](https://nodejs.org/), ideal for **large-scale, enterprise-grade** apps.
- Supports **modular architecture**, is easy to maintain, and scale.
- Built-in support for:
 - REST and GraphQL APIs
 - Authentication
 - Validation and serialization
- Compatible with **microservices**, if scalability is required in the future.

Compared to Express or Koa:

- NestJS provides **more structure and tooling out of the box**, reducing boilerplate and improving maintainability.

Benefits:

- **Client:** Faster backend development and more reliable output due to built-in conventions.
- **Maintenance:** Code is **organized by modules**, making it easier to scale, test, and debug.

11.4. PostgreSQL – *Database Technology*

Why PostgreSQL?

- A powerful **open-source RDBMS** used by companies like Apple, Instagram, and Reddit.
- Supports both:
 - **Structured (relational)** data via SQL
 - **Semi-structured (JSONB)** data when flexibility is needed
- Highly **scalable**, **ACID-compliant**, and includes built-in **full-text search, indexing, and geolocation** capabilities.

Compared to MySQL or MongoDB:

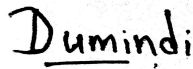
- More advanced features than MySQL (like JSON indexing).
- Combines the **strength of relational and NoSQL** in one system, better than using MongoDB unless purely document-based.

Benefits:

- **Client:** Secure, fast, and battle-tested database performance.
- **Maintenance:** Easier to query and maintain complex data with hybrid relational + unstructured design.

12. Declaration

We as members of the project titled OnSite360, certify that we will carry out this project according to guidelines provided by the coordinators and supervisors of the course as well as we will not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any university. To the best of our knowledge and belief, the project work will not contain any material previously published or written by another person or ourselves except where due reference is made in the text of appropriate places.

<i>Name</i>	<i>Signature</i>
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(ii) P.I.P.N.Pemarathna	
(iii) A.P.D.Lochana	
(iv) G.R.C.T.Gamlath	
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